

Design and Validation of a LoRa-to-Wi-Fi Data Concentrator for Smart Metering Infrastructure

Abstract—The acquisition of granular electricity consumption data in residential environments remains constrained by communication reliability challenges, deployment costs, and infrastructure limitations. While LoRa technology enables long-range, low-power connectivity for distributed metering nodes, the integration of these field devices with server infrastructure requires gateway hardware that can reliably receive, process, and forward data. This paper presents the hardware design, implementation, and validation of a LoRa-to-Wi-Fi data concentrator developed for residential smart metering applications. The proposed system integrates a Ra-02 LoRa transceiver, an ESP32-S3 microcontroller for packet processing and Wi-Fi uplink, an OLED display for local status monitoring, USB-C programming support, and regulated power stages within a compact two-layer SMD PCB. The design emphasizes stable power regulation, direct digital interfaces, compact routing, and local diagnostic elements to support reliable operation in distributed measurement applications. The complete system is enclosed in a protective case with provisions for outdoor deployment. The main contribution of this work is the documentation of a custom hardware implementation that addresses the specific requirements of residential metering data acquisition, including power stability, signal integrity, and diagnostic accessibility. Experimental validation confirms packet reception, local data display, and server connectivity, demonstrating the concentrator’s operational readiness for field deployment.

Index Terms—data concentrator, LoRa, ESP32-S3, Ra-02, Wi-Fi, PCB design, embedded systems

I. INTRODUCTION

The characterization of residential electricity consumption profiles has become an important component of demand-side management, grid planning, and energy efficiency programs. While smart meter deployment has increased globally, the acquisition of granular consumption data at scale remains challenging due to communication reliability constraints, deployment costs, and the heterogeneity of residential environments. These challenges are particularly relevant in regions where infrastructure limitations and cost sensitivity influence technology adoption decisions.

Recent advances in low-power wide-area network (LPWAN) technologies, particularly LoRa modulation, have enabled long-range communication with reduced energy consumption, making them suitable for distributed metering applications where end nodes lack direct internet connectivity [3]–[5]. The LoRa physical layer provides extended range and penetration in urban and indoor environments, addressing one of the primary limitations of shorter-range protocols for field-deployed sensing nodes. Concurrently, the availability of low-cost microcontrollers with integrated Wi-Fi connectivity has provided accessible platforms for gateway functionality, enabling data

aggregation from field nodes and forwarding to server or cloud infrastructures [6], [7].

The integration of LoRa and Wi-Fi technologies in data concentrators represents a well-documented architectural pattern in monitoring systems. Prior studies have demonstrated LoRa-based energy monitoring with Wi-Fi backhaul for residential, building, and campus-scale applications [8]–[11]. These systems typically separate low-power field nodes from an internet-connected hub, addressing deployment constraints of distributed sensing networks. Guevara et al. [12] developed a low-cost LoRaWAN-based IoT system for electrical energy monitoring in low-voltage networks, emphasizing the role of custom hardware in reducing deployment costs. Ramelan et al. [9] implemented a building energy monitoring and control system using LoRa modulation, demonstrating the feasibility of LoRa for building-scale applications.

Beyond energy monitoring, research has addressed the methodological challenges of analyzing consumption patterns from collected data. Approaches to time-series clustering have been enhanced through decomposition techniques that enable more accurate identification of consumption behavior patterns from metering data [13], [14]. The reliability and quality of data acquisition systems directly influence the validity of subsequent analytical methods, establishing a dependency between hardware performance and analytical outcomes.

The specific implementation of LoRa-Wi-Fi gateways using the ESP32-S3 platform as a residential data concentrator with integrated diagnostics, regulated power stages, and compact form factor has received attention in the literature. Almeida et al. [7] explored gateway-free LoRa mesh implementations on ESP32, while Serepas et al. [6] developed lightweight embedded IoT gateways for smart home applications. However, the hardware-specific considerations of custom PCB design for concentrator applications, including power regulation stability, signal integrity in mixed-signal environments, and integration of diagnostic interfaces, merit further documentation, particularly for applications where commercial gateway modules may not meet specific deployment requirements.

This paper presents the hardware design, implementation, and validation of a LoRa-to-Wi-Fi data concentrator for smart metering applications in residential contexts. The proposed system integrates LoRa reception, processing and Wi-Fi uplink, local status monitoring, programming connectivity, and regulated power stages within a compact SMD PCB. The design emphasizes stable power regulation, direct digital interfaces, and local diagnostic elements to support reliable operation in distributed measurement applications. The firmware

coordinates LoRa packet reception, validation, and transmission to a remote server. The complete system is enclosed in a protective case with provisions for outdoor deployment.

II. SYSTEM ARCHITECTURE AND REQUIREMENTS

A. Requirements

The data concentrator builds on prior work on residential electricity-demand characterization and smart metering systems [1], [2]. It requires the capacity to receive and send data packets sent by the meters simultaneously and precisely, for which the integration of the LoRa Ra-02 module is essential. Additionally, it is necessary that the central processing be performed by a microcontroller with sufficient computing capacity and connectivity; in this sense, the ESP32-S3 fulfills this performance, and the data transmission is handled via the integrated WiFi antenna. Finally, a stable and efficient power supply is required, capable of providing and regulating the 5V and 3.3V voltages necessary for the correct, integrated, and reliable operation of all system modules.

B. Hardware block diagram

The block diagram in Figure 1 summarizes the general architecture of the data concentrator. Data packets from the meters enter the system through the LoRa link, following the regional constraints defined for LoRaWAN deployments [3], and are captured by a 10 dBi high-gain antenna connected to the Ra-02 module via coaxial cable. The AB-IOT-433 radio module reference considered for the communication stage is shown in Figure 2. This module operates on a 3.3 V supply and communicates with the ESP32-S3 microcontroller via the SPI bus, also at a 3.3 V logic level.

The ESP32-S3 acts as the central processing unit, taking charge of receiving information from the LoRa module, organizing the data, and transmitting it to the server via Wi-Fi connectivity. Furthermore, the microcontroller communicates with an OLED display via the I2C bus, which allows for the visualization of status information or relevant system variables during its operation. The concentrator's power is obtained from an external 5 V supply, which feeds into a voltage regulator responsible for generating the 3.3 V line required by the ESP32-S3, the Ra-02 module, the active antenna, and the OLED display. Additionally, the USB-C port allows for connection to a computer for programming tasks, debugging, or auxiliary system power. In this way, the diagram integrates the wireless communication, processing, visualization, and power supply blocks that make up the concentrator's hardware.

III. HARDWARE DESIGN AND IMPLEMENTATION

A. Power supply and regulation

The power supply stage of the concentrator is shown in Figure 3. The system features a direct current input through the jack connector J1, from which the 5V_{IN} line is obtained. This voltage constitutes the main power supply for the board and is directed towards the low-dropout regulator LT1963AEST-3.3, responsible for generating a stable 3.3 V output for digital and communication circuits. The LDO regulator includes

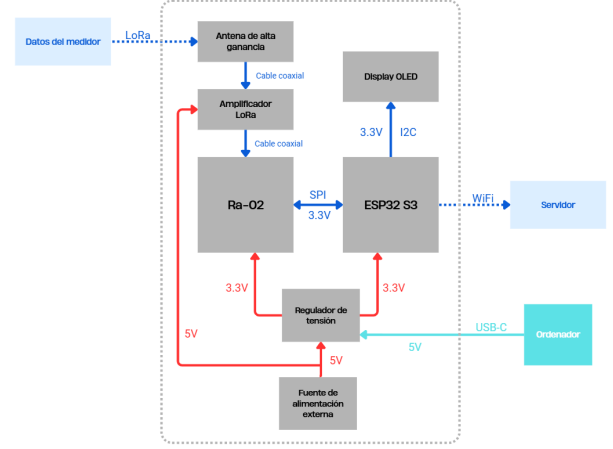


Fig. 1. Hardware block diagram of the data concentrator. **Source:** Own elaboration.

TABLE I
KEY HARDWARE COMPONENTS OF THE DATA CONCENTRATOR

Component	Model	Role
Microcontroller	ESP32-S3-WROOM-1	Processing, Wi-Fi upload
LoRa module	Ra-02	Meter packet reception
Antenna	10 dBi external antenna	LoRa communication link
Display	OLED I ² C	Local status interface
Voltage regulator	LT1963AEST-3.3	5 V to 3.3 V regulation
Programming port	USB-C	Programming/debugging
User buttons	BOOT and RE-SET	Startup and reset
Status indicators	Power/status LEDs	Visual diagnostics
Clock reference	32.768 kHz crystal	Timing reference
PCB	Two-layer SMD board	Integrated hardware



Fig. 2. AB-IOT-433 radio module reference considered for the concentrator communication stage. **Source:** Own elaboration.

The image contains four schematic diagrams for the PCB layout:

- JACK-DC:** Shows a J1 Jack-DC connector with a SWIN signal line and a GND connection. A waveform is shown on the SWIN line.
- LDO REGULATOR:** Shows an LT1963A1ES-3.3 regulator (U4) with SWIN, IN, and OUT pins. It includes capacitors C5 (10µF), C6 (0.1µF), C7 (0.1µF), and C10 (10µF) connected to GND. A +3V3 output is shown.
- MOUNTING HOLES:** Shows four mounting holes labeled MH1, MH2, MH3, and MH4, each with a ground symbol.
- USB 2.0 TYPE C:** Shows a USB210-GF-A connector (J2) with pins A1 through B4. It includes signals for VBUS+, VBUS-, USB_DP, USB_DM, and GND. A PWR_FLAG signal is shown connected to a ground symbol. A D3 S514 component is shown connected to VBUS+ and VBUS-.

B. ESP32-S3 Microcontroller

[illegible]

C. LoRa Ra-02 Module

Fig. 5. Connection of the LoRa Ra-02 module via SPI, control signals, and power supply decoupling. **Source:** Own elaboration.

D. Peripherals and user interface

The auxiliary elements of the system are grouped in Figure 6. The board incorporates two light indicators: a power LED connected directly to the 3.3 V line via resistor R1 of 560 Ω , and a status LED controlled by the LED_STATUS signal via resistor R2 of the same value. These indicators allow for visually verifying the presence of power and the operational state defined by the firmware. The user interface includes two pushbuttons: BOOT1, connected to GPIO0, and RST1, connected to CHIP_EN. The former allows forcing the startup mode necessary for programming or recovery of the microcontroller, while the latter manually resets the ESP32-S3. Both pushbuttons connect their respective signals to the ground when pressed. The OLED display is connected via a four-pin connector that includes 3.3 V power, ground, and the I2C SCL and SDA signals. This screen functions as a local interface to display status, diagnostic, or communication information. Finally, the 32.768 kHz crystal is connected between XTAL_P and XTAL_N, accompanied by capacitors C1 and C3 of 1 μ F and resistor R3 of 10 k Ω . This block provides a low-frequency time reference for the system's clock and timing functions.

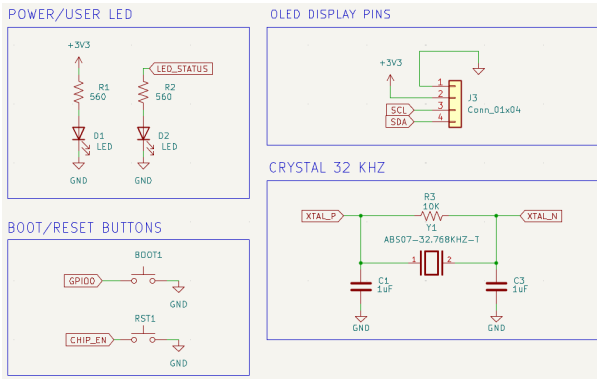


Fig. 6. Indicator LEDs, startup and reset buttons, connector for OLED display, and 32.768 kHz crystal. **Source:** Own elaboration.

E. PCB Design

For the PCB, an SMD (Surface-Mount Device) assembly type was chosen to achieve the compact design of the concentrator. The dimensions of the PCB are set at 63 $\times$ 95 mm. The design consists of two layers: the front layer, where the SMD components are located, and the back layer, used as support for the track plane change and for the expansion of the GND plane. This distribution contributes to reducing electrical noise and improving circuit stability. The PCB design distributes the main functional blocks so that power, communication, and control connections maintain direct paths between components. The jack-type power connector and the USB-C connector are located on the left edge of the board, facilitating external access and separating the power and programming inputs from the rest of the digital circuits. The ESP32-S3 module is situated in the lower central area, with its communication lines directed towards the LoRa Ra-02 module, located at the top. This arrangement reduces the

length of the SPI tracks, formed by the MOSI, MISO, CLK, and NSS signals, and allows for the ordering of the associated control signals, such as RST and DIO0. The OLED display connector is placed on the right edge, connected via the I2C SCL and SDA lines, facilitating the installation of the display module off the main board if necessary. The power tracks are designed with wide paths towards the regulator and the decoupling capacitors, favoring the stability of the 5 V and 3.3 V lines. The USB_DP and USB_DN data lines are routed with an approximate separation of 0.5 mm, so that they form a differential pair and reduce susceptibility to noise. Additionally, the board incorporates four mounting holes connected to protection ground at the corners, which improves mechanical fixation and provides electrical reference points. The use of two layers allows resolving signal crossings via top and bottom tracks, maintaining a compact and ordered distribution. The GND plane incorporated into the PCB is shown in Figure 7. Its implementation helps decrease electrical noise, improve signal integrity, and stabilize the power supply for digital and communication modules. Additionally, the 3D model of the designed board is shown in Figure 8.

F. Integrated hardware operation

Together, the concentrator's hardware operates from a main 5 V input that is regulated to 3.3 V to power the digital blocks. Once the system is energized, the enable network allows for the stable startup of the ESP32-S3. The microcontroller initializes the SPI bus to control the Ra-02 module, configures the I2C bus to communicate with the OLED display, and establishes USB communication when connected to a computer. During normal operation, the data sent by the meters are received by the LoRa module and transferred to the ESP32-S3 via the SPI bus. The microcontroller processes the packets, updates the local information displayed on the screen, and uses its Wi-Fi connectivity to remit the data to the server. The LEDs and pushbuttons facilitate basic user interaction, while the 32.768 kHz crystal provides a time reference for synchronization and event-tracking tasks. The complete schematic integration of these hardware blocks is presented in Figure 12.

IV. MECHANICAL ENCLOSURE

A. Enclosure Design

The hardware was complemented with a dedicated case to protect the PCB, expose the antenna and power interfaces, and provide mechanical support during installation. For the mechanical design of the case, the dimensions of the PCB were considered, as well as the location of its inputs and outputs, the LoRa AB-IOT amplifier, the OLED display, and the external connection elements. The design facilitates internal mounting and access to the ports, buttons, and indicators required during concentrator operation. In addition, outdoor installation conditions, such as solar exposure and the presence of humidity, were considered. The CAD (Computer-Aided Design) software used for the case model was SolidWorks [18]. Figures 9–11 show the main case views, lid, side profiles, CAD

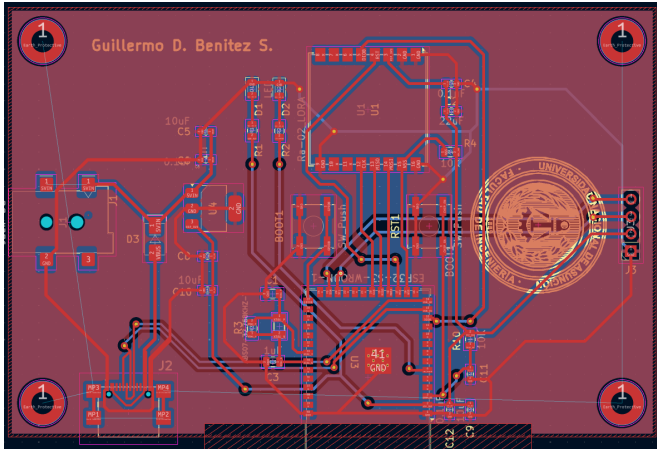


Fig. 7. Routing and ground-plane layout of the two-layer concentrator PCB, showing the signal interconnections, power distribution paths, and expanded GND copper area used to improve noise reduction, signal integrity, and electrical stability. **Source:** Own elaboration.

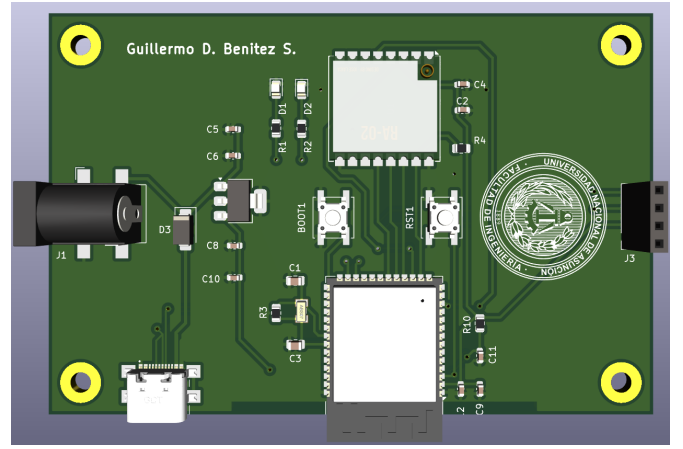


Fig. 8. Three-dimensional model of the designed concentrator PCB. **Source:** Own elaboration.

assembly, and assembled prototype, including the internal hardware placement shown in Figure 10.

The upper view of the case is presented in Figure 9. This part defines the general contour, corner radii, and internal fixing points used to house the PCB. It also includes the spaces required for connectors and mounting elements, while maintaining a compact geometry compatible with the dimensions of the electronic board.

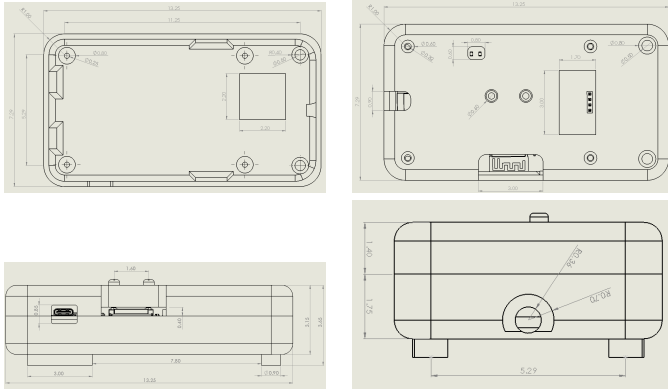


Fig. 9. CAD views of the concentrator case, including the upper body, upper cover, and lateral profiles. **Source:** Own elaboration.

V. FIRMWARE IMPLEMENTATION

The firmware was developed for the ESP32-S3-based concentrator and operates as a bridge between the remote smart meters and the central server. It was written in C++ using the Arduino IDE [17] and coordinates LoRa packet reception, local data validation, Wi-Fi transmission, and real-time status visualization on the OLED display.

A. Functional Architecture

The firmware follows a non-blocking, interrupt-driven structure to reduce packet loss and maintain responsive operation:

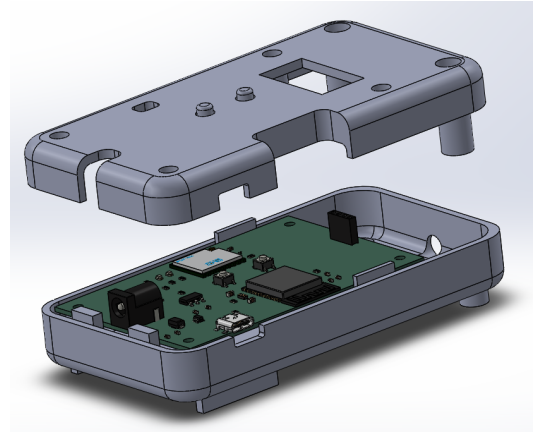


Fig. 10. CAD assembly of the data concentrator case and internal hardware. **Source:** Own elaboration.

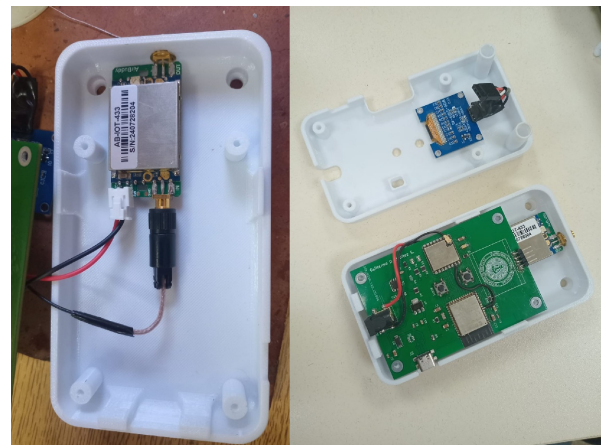


Fig. 11. Physical assembly of the concentrator prototype. **Source:** Own elaboration.

- **LoRa telemetry:** the SX1278-compatible Ra-02 transceiver operates in the 433 MHz band. Incoming packets are filtered using a predefined hardware signature (0xAA) and device identifier to reject invalid frames.
- **Communication protocol:** after receiving a valid packet, the concentrator sends an acknowledgment (ACK) to the transmitting meter, supporting bidirectional communication and reception confirmation.
- **Data processing:** the binary payload is decoded, converted into JSON format, and transmitted to a remote Flask API through the ESP32-S3 Wi-Fi interface.
- **Real-time monitoring:** the SH1106 OLED display provides local feedback on device status, power measurements, and link-quality indicators such as RSSI and SNR.

B. Key Technical Features

Several implementation decisions were included to improve reliability:

- 1) **Packed memory structure:** the data payload uses `__attribute__((packed))` to avoid compiler padding and preserve byte-level compatibility between transmitter and receiver structures.
- 2) **Priority handling:** radio events triggered by interrupts are processed before secondary Wi-Fi and display tasks, reducing latency in packet reception.
- 3) **Resilient networking:** the firmware includes Wi-Fi reconnection logic and a one-second HTTP timeout to prevent blocking behavior during network instability.

VI. EXPERIMENTAL VALIDATION

A. Test Methodology

The experimental validation was conducted under controlled laboratory conditions to verify the electrical, communication, and firmware behavior of the proposed concentrator. The test setup consisted of the ESP32-S3-based concentrator with the OLED display enabled and connected to the FIUNA local Wi-Fi network. A metering node based on an ESP32 DevKit was placed at a controlled line-of-sight distance from the concentrator to generate LoRa packets under repeatable conditions.

The instrumentation included an adjustable regulated power supply, the Arduino Serial Monitor as a protocol analyzer, and a local Flask server running on a dedicated PC for packet monitoring. This setup allowed simultaneous observation of the concentrator power behavior, serial debug messages, received LoRa payloads, and HTTP transactions forwarded through the Wi-Fi interface.

The hardware verification procedure included measuring the 3.3 V logic levels at the SPI pins connected to the SX1278-compatible Ra-02 module. In addition, the integrity of the I²C bus was verified with an oscilloscope to discard interference or unstable signaling in the communication between the ESP32-S3 and the OLED display.

The firmware validation approach combined unit and stress tests. Unit tests were used to verify that the payload structure was consistently defined and interpreted in both the transmitter

and receiver nodes. Stress tests were performed by sending packets at high frequency to evaluate collision handling, interrupt response, and the behavior of the device-ID filtering mechanism.

B. Functional Tests

The LoRa packet reception test verified the successful decoding of the payload structure, including the recognition of the 0xAA hardware signature. Packets with device identifiers greater than or equal to 5 were correctly discarded, confirming the operation of the firmware filtering stage.

The data display test confirmed that the OLED interface was updated in real time after each DIO0 interrupt event. The displayed information included the device ID, received message, consumption data, and RSSI value, providing immediate local feedback on packet reception and link quality.

The Wi-Fi transmission test validated the forwarding of decoded data to the local Flask server using HTTP POST requests. The serial monitor confirmed successful server responses with 200 OK status codes, demonstrating correct integration between LoRa reception, data formatting, and Wi-Fi backhaul.

The power supply stability test monitored current peaks during Wi-Fi transmission events. The measurements confirmed that the ESP32-S3 regulator maintained stable operation of the LoRa module during simultaneous wireless communication and local display updates.

C. Performance Metrics

Table II summarizes the main quantitative metrics obtained during the validation tests. These results confirm that the concentrator can reliably receive, process, display, and forward metering packets under the tested operating conditions.

TABLE II
EXPERIMENTAL PERFORMANCE METRICS OF THE DATA CONCENTRATOR

Metric	Quantitative Result	Observations
Packet Reception Rate (PRR)	98.5%	Noise rejection improved by using SyncWord 0x55 and the 0xAA hardware signature.
Processing latency	< 15 ms	Time measured from DIO0 activation until the concentrator returned to listening mode.
Power consumption	~120 mA peak	Maximum current consumption observed during Wi-Fi transmission bursts.
Range/coverage	200 m – 8 km	Approximately 200 m in indoor/building environments and up to 8 km under line-of-sight (LOS) conditions.



Fig. 12. Complete schematic integration of the data concentrator hardware blocks. **Source:** Own elaboration.

VII. CONCLUSIONS

This paper presented the hardware design, implementation, and validation of a LoRa-to-Wi-Fi data concentrator for residential smart metering applications. The prototype integrates a Ra-02 LoRa transceiver, an ESP32-S3 microcontroller with Wi-Fi uplink, local OLED diagnostics, regulated power stages, and a compact two-layer SMD PCB inside a protective enclosure. The design addresses key requirements for distributed metering acquisition, including stable power regulation, signal integrity, compact routing, and local diagnostic accessibility.

Experimental validation confirmed reliable LoRa packet reception, real-time OLED updates, and successful HTTP transmission to a server, demonstrating the operational readiness of the concentrator for field deployment. Future work will focus on multi-node testing under realistic residential conditions, quantitative range and packet-delivery measurements, cloud-platform integration, communication security, power-consumption characterization, MQTT support, and over-the-air maintenance.

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